Md Nazmul Islam Ananto

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Education

The Pennsylvania State University **Aug 2026 – Present**
PhD in Computer Science and Engineering *University Park, PA, USA*

Bangladesh University of Engineering and Technology (BUET) **Apr 2019 – Jul 2024**
BSc in Computer Science and Engineering *Dhaka, Bangladesh*

- **CGPA: 3.91** ; Dean's List Award
- Notable Courses: Machine Learning, Compiler, Computer Networking, Security, AI, Architecture, Operating Systems, Database, Data Structure & Algorithm

Research

ToFu: Exploiting and Fuzzing the LLM-Orchestrator–Tool Trust Boundary
[Preprint] — submitted to FSE'27

- A semantics-aware fuzzing framework for LLM agent tools that derives security checks from tool metadata and a corpus of 74 CWE-guided sanitizers
- Found 253 manually verified vulnerabilities across 118 open-source MCP servers, including path traversal, SSRF, command injection, log injection, and data exfiltration, none of which crash-based fuzzers detect

CompassLLM: A Multi-Agent Approach toward Geo-Spatial Reasoning for Popular Path Query
*[**ACL'26** Findings] — supervised by Mohammed Eunus Ali & Md Rizwan Parvez*

- We presented a multi-agent framework to tackle geo-spatial reasoning problems, particularly the popular path query, accommodating both searching and generating paths from historical trajectory data
- Evaluated using GPT-4o, GPT-o3-mini and Llama 3.1 8b and demonstrated superior performance on real and synthetic datasets, especially in sparse data scenarios where other baselines underperform

Experience

Research Assistant **Aug 2026 – Present**
PSecLab @ The Pennsylvania State University *University Park, PA, USA*

- Working on AI Security, Agentic Systems and LLM Manipulation
- Supervised by Dr Arslan Khan

Adjunct Lecturer **Nov 2025 – Jun 2026**
Bangladesh University of Engineering and Technology (BUET) *Dhaka, Bangladesh*

- Conducting classes on Networking, Signal Processing and Digital Logic Design

Software Engineer **Mar 2024 – Jul 2026**
Zelf AI *Brooklyn, NY, USA*

- Building data pipelines collecting and processing 80M+ public videos from social medias
- Developing automation tools serving 100+ enterprise clients. Collaborating with retail partners and cross-functional teams for critical features
- Building AI pipelines to elicit audio-visual insight from social media videos and generate analysis on them
- **Tech Stack**: Python, GPT, Gemini, FastAPI, Selenium, Playwright, Appium, Burp Suite, Mitmproxy, ADB, GCP, Docker, GitHub Actions, Terraform

Software Consultant **Jun – Aug 2025**
Shottify *Los Angeles, CA, USA*

- Conducted sessions on Information Retrieval and Social Media Analysis
- Developed News Aggregator for renowned international news portals including NYT, Reuters - [see more]

Research Assistant **Jul 2022 – Jun 2024**
Institute of Water and Flood Management, BUET *Dhaka, Bangladesh*

- GIS Data Visualization for river erosion prone areas using Google Maps, React & Express - [live here]
- Published in *Frontiers* [paper]

Scores & Achievements

1st / 400 – Zelf Hackathon 1.0 **Feb 2024**
Built an automation platform that solved long-standing engineering backlogs; led to recruitment interview
5th / 68 – IEEE GameJam [play now] **Jan 2023**
BUET Technical Scholarship **Apr 2019**
HSC & SSC Scholarship **2018 & 2016**

Projects

Online News Scraper | *Python, FastAPI*

- News Aggregator for renowned international news portals including NYT, Reuters

Stock Trader | *ML, Python*

- Trained a rule-based stock trading agent on DSE time-series data; yielded 18% ROI in simulated backtests

Maps 'n Bags | *ExpressJS, PostgreSQL, Firebase, Selenium*

- Developed an itinerary planner with custom trip builders, suggestions, travel tracking & budgeting with synced data & distances via Google Maps API

Creative Production Management | *System Design, BPMN, UI Mockups*

- Designed an end-to-end production workflow for a creative agency — covering quotation, delegation, man-hour tracking, and final delivery

Withered Away | *ExpressJS, OracleDB*

- Built dashboard to manage residents, amenities, and room allocations for retirement home officials, doctors, and members

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, C++, Bash, x86 Assembly
Full-stack: Flask, FastAPI, React, Express, NestJS, Apollo, Prisma
DevOps: GCP, Git, Docker, Kubernetes, Terraform, Github Actions, Firebase, Linux Shell
Testing & Automation: Selenium, Playwright, Appium, Burp, ADB, Reverse-engineering
Libraries & ML: NumPy, Pandas, LangChain, OpenCV, PyTorch, scikit-learn, NLTK, Matplotlib, Seaborn
Databases: PostgreSQL, MongoDB, MySQL, Oracle

Extracurricular

Director of Marketing & Communication **Jun 2023 – Apr 2024**
BUET Cyber Security Club *Dhaka, Bangladesh*

- Promoted cybersecurity events & hosted inter-university CTF (Catch The Flag) contests

Organizer **Jun – Jul 2023**
BUET CSE Fest 2023 *Dhaka, Bangladesh*

- A proud part of the organizing team of the 2023 edition of the annual departmental festival consisting inter-university competitions such as hackathon, programming contest, deep learning competition, CTF competition and extra-curricular events such as cultural program, dance mob, treasure hunt, indoor and outdoor games

President of Writing & Publication **Feb 2018 – Feb 2019**
Notre Dame English Club *Dhaka, Bangladesh*

- Organized multiple national level English Carnivals along with taking frequent sessions on English learning; published three magazines as an editor and multiple wall-magazines